

Cover sheet for submission of work for assessment



UNIT DETAILS

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Executive Summary

A thorough examination of SparkEV's statistics from 2019 to 2023 provided important insights and suggestions that are essential for the business continuing success in the electric vehicle (EV) market. SparkEV, a minor participant in the automotive manufacturing industry, encounters data management challenges, as do all data-reliant companies. Ensuring high-quality data integrity, addressing privacy and security risks, and maintaining reliable decision-making processes are essential. The utilization of data-cleaning techniques enhances the accuracy and practicality of the analysis.

The report highlights the importance of SparkEV's robust market position in the expanding North America. The business should strengthen its marketing initiatives, forge solid local alliances, and improve customer service in order to solidify its position. Simultaneously, SparkEV ought to broaden its scope in Europe and Asia, customizing its approaches to suit these regions' particular needs. Nonetheless, these markets should follow the best practices that have propelled the observed growth in North America.

The importance of adaptability in response to market changes is emphasized by financial analysis. Adapting pricing tactics to certain market niches might maximize financial gains. Operational insights recommend reframing Tempest, the most profitable model, to better fit with pricing and market appeal, streamlining the product range, and learning from the Centro model's operational effectiveness. Expanding the product line, carrying out in-depth market analysis, and putting cost control measures in place are essential for reaching a wider audience and increasing revenue. The paper also emphasizes the importance of improve sales and distribution channels by bolstering the dealer network, boosting the retail experience, and aligning with regional preferences.

To summaries, SparkEV's entry into the electric car market necessitates an agile strategy that includes financial management, production, marketing, and product optimization. Ongoing investments in R&D, thorough market research, and partnerships with local partners are crucial. Data management, financial analysis, performance analytics, operational insights, forecasts, and practical recommendations are all included in the analysis, which aims to position SparkEV for sustainable industry leadership.

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I. Introduction

1. Background

The automotive industry, historically known for its innovations, is witnessing a transformative era with the rise of Electric Vehicles (EVs). As global efforts intensify to combat climate change and reduce carbon footprints, EVs emerge as a beacon of sustainability in transportation (Sanguesa et al., 2021).

SparkEV, positioned at the forefront of this movement, exemplifies the blend of technological prowess and visionary leadership needed in today's EV landscape. According to Sun et al. (2019), the shift from fossil fuels to electric propulsion is not just an environmental imperative but also an economic one. Companies like SparkEV stand at a pivotal juncture, ready to shape the future of mobility.

However, the journey for SparkEV has its challenges. Technological advancements, infrastructural demands, consumer preferences, and market dynamics constantly evolve, necessitating a proactive and informed approach. Therefore, understanding these dynamics is crucial for any EV company aiming to maintain a competitive edge (Sun et al., 2019).

2. Purpose and outline

This report endeavors to dissect these multifaceted aspects, providing a holistic analysis tailored for SparkEV. We will delve into performance, financial, and operational analytics to highlight key trends, strengths, and areas for optimization. Using historical data, we will also project a two-year forecast, arming SparkEV's management with actionable foresight vital for strategic planning and future competitiveness in the automotive landscape. Finally, drawing upon academic insights and industry trends, we aim to offer a comprehensive roadmap that can guide SparkEV through the intricate maze of the EV industry, ensuring its continued success and growth.

3. Limitations

The report has limitations in terms of data quality and scope. The quality is not guaranteed as it is a provided sample dataset, and the team lacks control over its collection, potentially affecting

analysis reliability. Data cleaning uses only Tableau Builder Prep, addressing some quality issues, but further statistical analysis falls outside the project's scope. Additionally, the data's limited scope covers only four years and four specific regions, possibly inadequate for identifying long-term trends or representing SparkEV's entire global operations.

II. Data Management

1. Data utilization in business

With the rise of innovative IT capabilities, particularly big data analytics, there is significant potential for improving firm performance and exploring new business opportunities (Kwon et al., 2014). However, to effectively utilize analytics, meticulous data handling is essential. Data collection and preparation, including data cleaning for quality and integrity, serve as the initial steps (Gandomi & Haider, 2015). Quality data, as highlighted by Kwon et al. (2014), is crucial for insightful decision-making and encompasses attributes such as ***accuracy, completeness, consistency, timeliness, and accessibility***. Notably, a firm's capacity to maintain the quality of its corporate data is a pivotal element contributing to data integrity. Therefore, the journey of leveraging big data analytics begins with the foundation of high-quality data, and is fostered by data cleaning, an intricate process involving error identification, deduplication, schema-matching, and various other relational data issues (Krishnan et al., 2016). By taking steps to ensure data integrity, businesses can improve their decision-making, reduce costs, and avoid several problems.

2. SparkEV sample data

For this project, data collection is not performed. The focus will instead be on data cleaning, visualization, and analysis. The dataset contains two car models named Storm and Stormz, which will be treated as separate models rather than as accuracy or consistency issues.

Table 1. Dataset quality issues and implemented solutions.

Field	Changes made	Quality issue
VIN#	Removed 4 rows with duplicated VIN #6SEGW6E34DY634576 #6SEGW2E34DY465926	Accuracy: Duplicate data is a common problem in big data, as data is often collected from multiple sources. This can lead to errors in analysis and reporting.
Transmission	Split into Transmission - Speed and Transmission - Auto/manual	Consistency: Inconsistent data formats can make it difficult to analyze data. Splitting the Transmission field into two separate fields simplifies each data value and makes it easier to work with.
Country of Sale	Replaced “United States” by “USA” for uniformity	Consistency: Inconsistent data values can also make it difficult to analyze data. Replacing all instances of “United States” with “USA” makes the data more consistent and easier to compare.
Net Sales	Recalculated based on data definition = [Gross Sales] - [Variable Marketing]	Accuracy: Incorrect data calculations can lead to inaccurate results and wasted resources.
Contribution Margin	Recalculated based on data definition = [Net Sales] - [Total Variable Cost]	
Country names	Corrected spelling in “Philippines” Replaced “UK” by “United Kingdom” for uniformity	Accuracy: Misspelled data value, in this case, can cause inaccurate results. It occurs for a variety of reasons, such as human error, data entry errors, and optical character recognition (OCR) errors.
Tariff	Calculated based on data definition = [Gross Sales] - [Tariff Rate]	Completeness: Missing data values result in reduced statistical power and inaccurate conclusions. Calculating Tariff based on the correct data definition ensures that the data is complete and accurate.
Net Revenue	Recalculated based on data definition = [Contribution Margin] - [Other Costs]	Accuracy: Incorrect data calculations can lead to inaccurate results and wasted resources.
After-tax	Calculated based on data definition = [Net Revenue] - [Tariff]	

All data quality issues affect **timeliness** as they make it more difficult and time-consuming to collect, process, and analyze the data. **Accessibility** is not mentioned specifically, but is assumed, as the analysis and visualizations will be delivered to SparkEV. The company itself must have internal communication and resources in place to use the data.

3. Other critical issues

Big data offers significant potential benefits, including the ability to process vast amounts of data for analysis and decision-making (Herschel et al., 2017). However, it also presents risks for organizations like SparkEV. Big data can be inaccurate and incomplete due to its indiscriminate nature, potentially collecting unintended information and violating privacy (Fan et al., 2014). Privacy risks arise from reidentification when combining anonymized data sets, necessitating clear rules for data merging. Moreover, big data may lack accuracy, be biased, or lack context, leading to erroneous conclusions about customer behavior (Franzén, 2023). Security is a concern, with centralized big data structures posing new challenges that require safeguarding data confidentiality, integrity, and availability. SparkEV must address these issues for effective and secure big data utilization (Fan et al., 2014).

III. Industry Overview

The EV sector is witnessing an unprecedented surge in demand and adoption across the globe. With increasing environmental concerns and advancements in battery technology, EVs are quickly becoming a viable and often preferred alternative to traditional internal combustion engine vehicles. According to the International Energy Agency (IEA), the number of electric cars on the road skyrocketed from 17,000 in 2010 to 7.2 million in 2019, with 47% of these vehicles being in China. This marks a significant compound annual growth rate over this period (IEA 2020).

Around the world, governments are rolling out attractive incentives to accelerate EV adoption. Many have even planned out ambitious roadmaps for a full-scale shift to electric transport. For instance, the European Union aims for a substantial 37.5% reduction in CO₂ emissions by 2030 (IEA 2020). Such policy initiatives are catalyzing automakers to diversify and expand their EV portfolios (Bakker & Jacob Trip, 2013).

Driven by the urgency to reduce carbon emissions and mitigate climate change around the globe, major automobile players are entering the market with innovative and unique propositions (Ehsani et al., 2021). The growth of charging infrastructure, advancements in battery longevity, and the gradual reduction in costs are collectively fueling the sector's momentum. As of 2019, there were approximately 862,118 public charging points for EV globally. This number continues to grow as the infrastructure expands to meet new demand (IEA 2020).

Amidst this booming industry is SparkEV, a promising name in the EV segment. At present, SparkEV has successfully penetrated the market in multiple countries across Asia, Europe, and North America. Within these regions, they have marked a presence in several countries, ranging from mature EV markets to nascent ones. The brand showcases an eclectic mix of models tailored to cater to diverse segments of the population.

Our deep dive into SparkEV's dataset seeks to unravel the company's performance indicators, grasp prevailing market currents, and highlight potential enhancements. For an agile brand like SparkEV, sophisticated data analytics tools are indispensable (Awan et al., 2021). These tools vividly illustrate intricate aspects like sales distribution patterns, profit markers, and customer demographics.

IV. Analysis of key operations

A. Overall Performance Analysis

Overall performance analysis is a foundational tool for organizations to understand, assess, and improve their operations, strategy, and goal achievement (Matzler et al., 2004). By examining high-level performance metrics, insights into the popularity of product offerings, distribution channels, evolving trends, and regional performance can be gained.

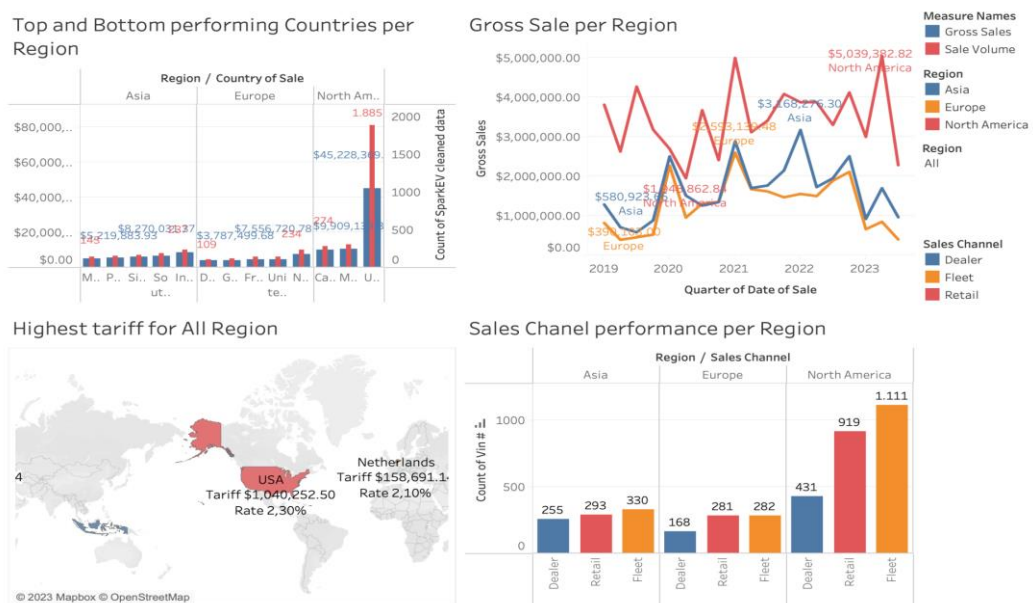


Figure 1.1. SparkEV Overall performance dashboard

1. SparkEV's global performance and market share

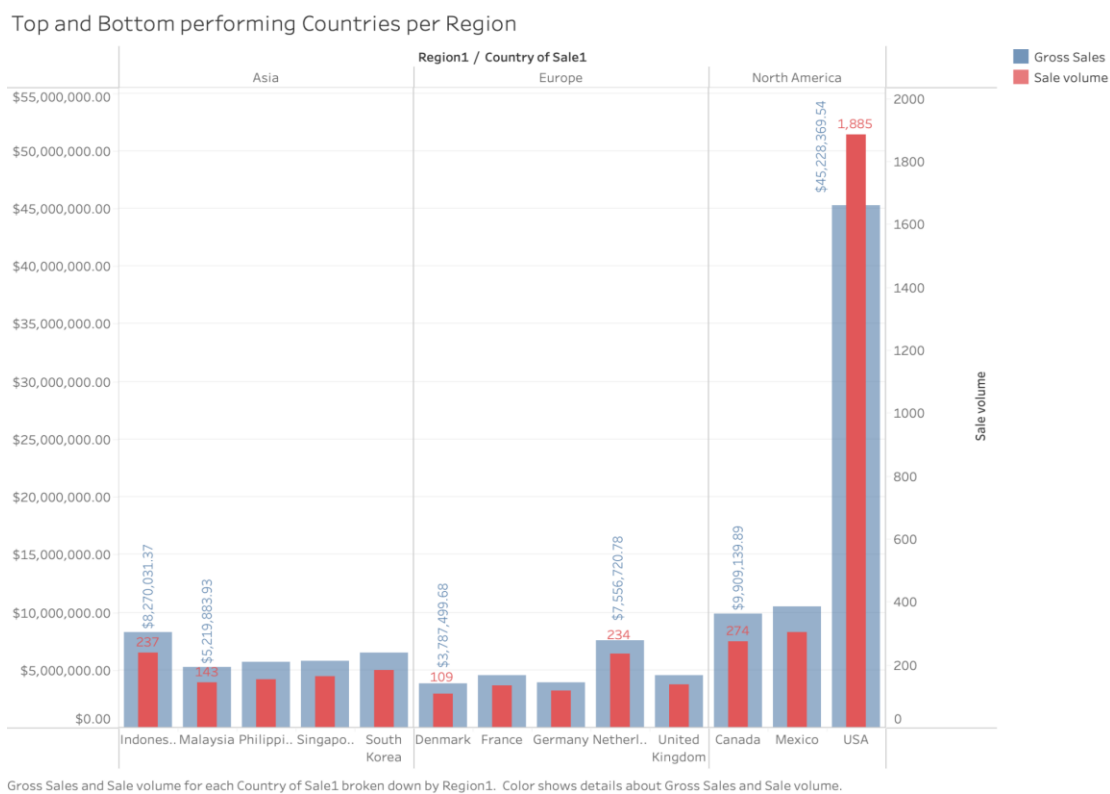


Figure 1.2. Top and Bottom performing Countries per Region

The analysis of SparkeV's global market performance reveals a regional disparity, with North America as the primary market, led by the United States with over \$45 million in gross sales. However, the US' upward trajectory (Electric Vehicles: Market Data Analysis & Forecast, 2022), SparkeV must navigate a highly competitive landscape dominated by major manufacturers, such as Tesla, Chevrolet, Ford, and others (ELECTRIC VEHICLE SALES REPORT — Q2 2023, 2023), leaving a tight space for market expansion.

In contrast, Denmark represents the smallest market for SparkeV at \$3.8 million. EVs, in fact, constitutes a mere 0.3% of the total registered vehicles despite government incentives. (Thøgersen & Ebsen, 2019). The largest consumer of Europe, the Netherlands, generates about double the amount. Europe's overall EV market remains constrained, with a low share of electric and hybrid vehicles in total car sales - comprising only 2.9% of the European Union's total car sales in 2015 (Ortar & Ryghaug, 2019).

Indonesia emerges as SparkeV's most prominent Asian customer, comparable to Europe's top consumer, underlining its growth potential. Notably, SparkeV has yet to enter the Chinese market despite its substantial consumption figures, indicating a significant untapped opportunity (IEA, 2023; Electric Vehicles: Market Data Analysis & Forecast, 2022)

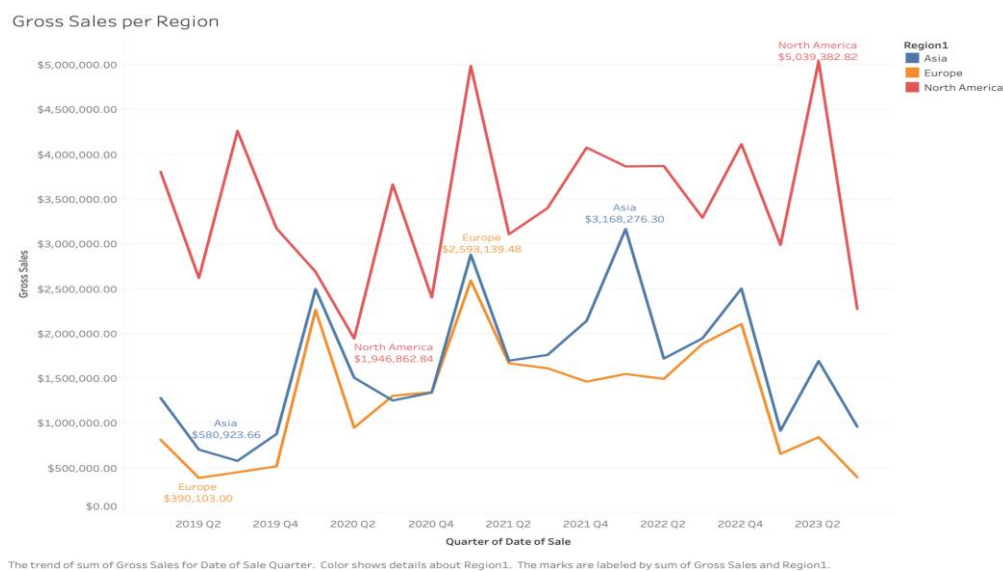


Figure 1.3. Gross Sales Trends per Region

The analysis of gross sales performance over the past five years demonstrates significant fluctuations in all three regions, sharing a common performance pattern. In Q3 2023, there was a decline across all regions, mainly due to the global chip shortage, rising material costs, and geopolitical issues (IEA, 2023). Notably, North America experienced spikes in sales during Q3 2019 and Q3 2020, while the other regions faced challenges.

Interestingly, from Q4 2019 to Q4 2020, North America's sales saw consecutive drops, in contrast to the surges in Europe and Asia. Besides more favorable incentives, Europe and Asia's resilience during COVID-19 can be attributed to regional supply chain differences and market adaptability (Gorner & Paoli, 2021). According to the same study, an upward trend in EV preference is persistent despite the pandemic, and a substantial peak was observed in Q1 2021. Further statistical analysis of SparkEV and the industry's supply chain can provide valuable insights for effective crisis management strategies.

2. Tariffs and government incentives

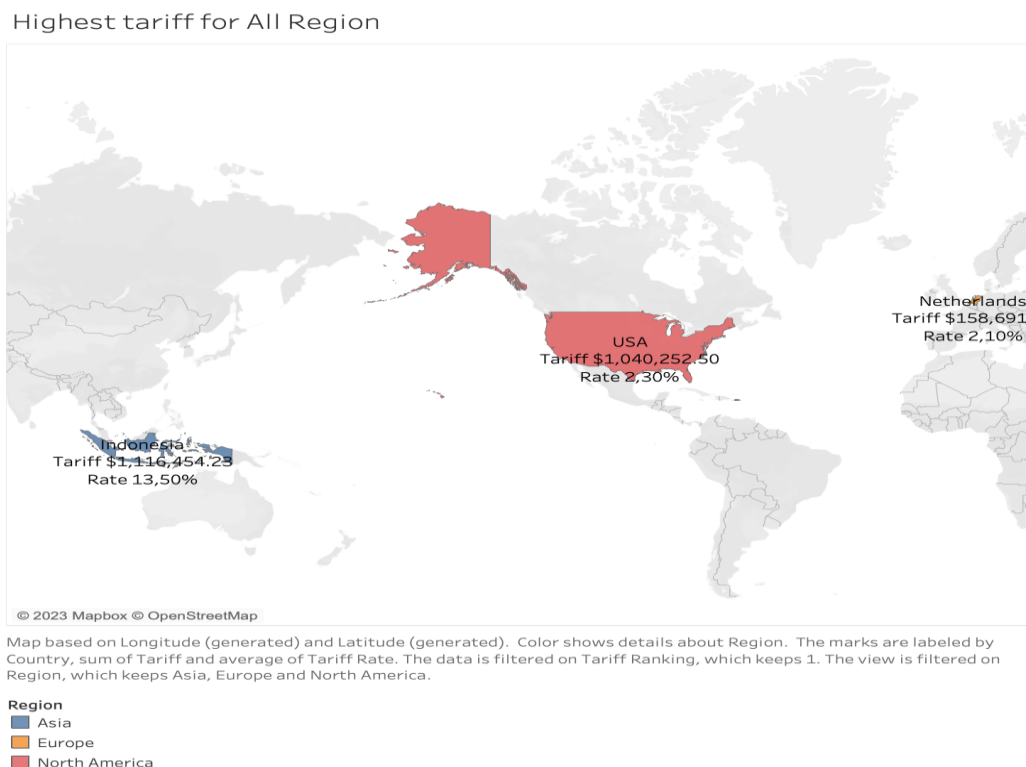


Figure 1.4. Top Country Tariff per Region

There are significant discrepancies in electric car tariffs among the regions. Indonesia has the highest average tariff at 13.5%, while the figures for the USA and the Netherlands are substantially lower at 2.3% and 2.1%, respectively. With substantially large sale volumes and relatively low tariff rates, the US market is highly favorable.

The high tariff in Indonesia was a significant barrier to export. However, since March 2022, the import tariffs on EVs have been drastically reduced from 24.1% to 0% (Global Trade Alert, 2022). This is a compelling opportunity to enhance its presence in the Indonesian market. Management should proactively focus on boosting Indonesian sales while the incentive policy lasts to maximize after-tax revenue and take full advantage of the favorable tariff environment (Setiawan et al., 2022).

3. Sales channels performance and customer demographic

Sales channels consistently demonstrate a uniform distribution of sales contribution for SparkEV across regions. The fleet option consistently outperforms the others, with retail closely trailing, while dealer purchases are the least favored choice. Notably, in North America, the fleet option accounts for slightly over half of the total retail sales volume. The robust performance of the fleet option could be attributed to corporate and business customers' inclinations for bulk purchases (Nesbitt & Sperling, 2001). Therefore, to leverage the popularity of the fleet option, enhance the retail experience, and diversify sales channels, SparkEV should focus on improving the dealer network, ultimately maximizing sales contribution.

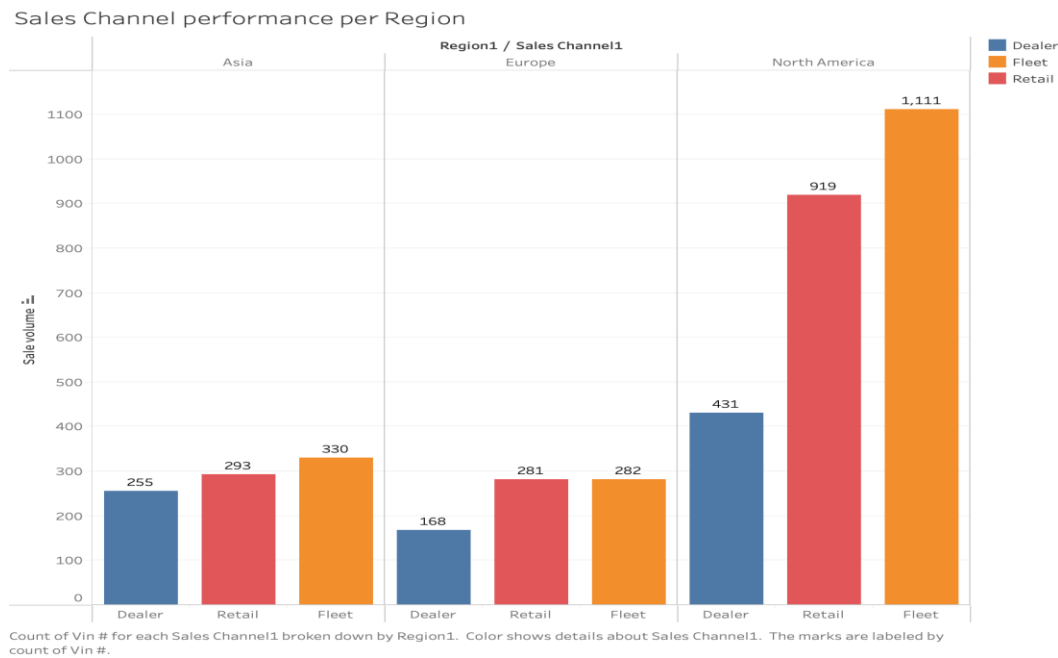


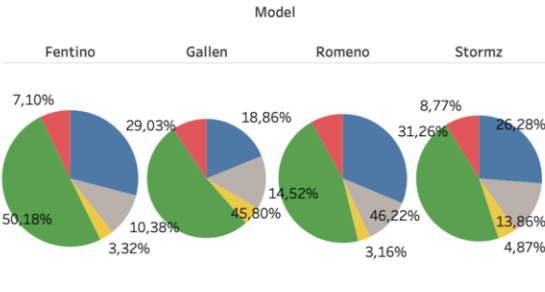
Figure 1.5. Sales Channel Performance per Region

In 2022, fleet sales accounted for approximately 30% of total vehicle sales in North America, while retail accounts for about 82% of new registrations in North America and 41% in Europe (Pfiefe et al., 2018). SparkEV's sales segmentation is, therefore, disproportionate to the industry average, having a higher fleet and lower retail sales.

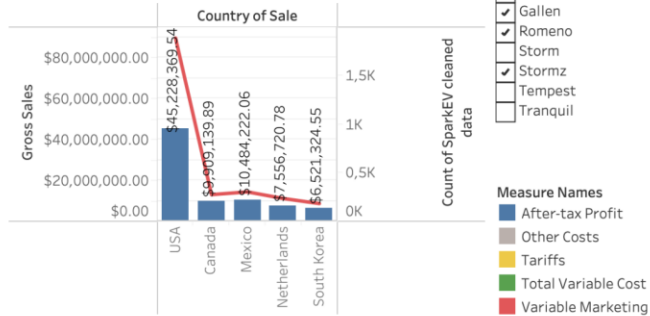
Fleet sales consist of large corporations, firms, and government agencies and expanded to small business automotive companies (Zeimpekis & Tarantilis, 2007). Like other automotive companies, SparkEV's emphasis is on Business-to-Business (B2B) sales, and the core customer segment does not divert from the norm for automotive companies (Chirumalla, 2023). The success of fleet sales across different regions confirms a robust and well-resonated B2B strategy for a diverse range of business customers. The company has identified and capitalized within the corporate and institutional sectors, establishing a strong position in the B2B market.

B. Financial Analytics

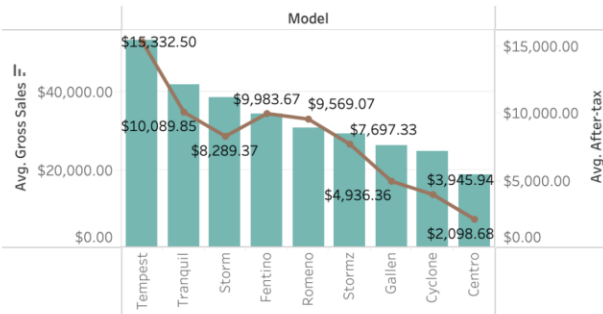
Gross sale components of Fentino, Gallen, Romeno and 1 more model(s)



Top 5 countries based on Net Revenue and theirs Sale Volumes



Average Gross Sale vs Average After-tax profit by models



Total Quarterly Gross Sales vs Net Revenue by All region(s)

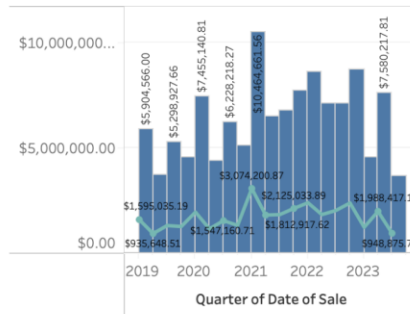


Figure 2.1. SparkEV Financial Analytics dashboard

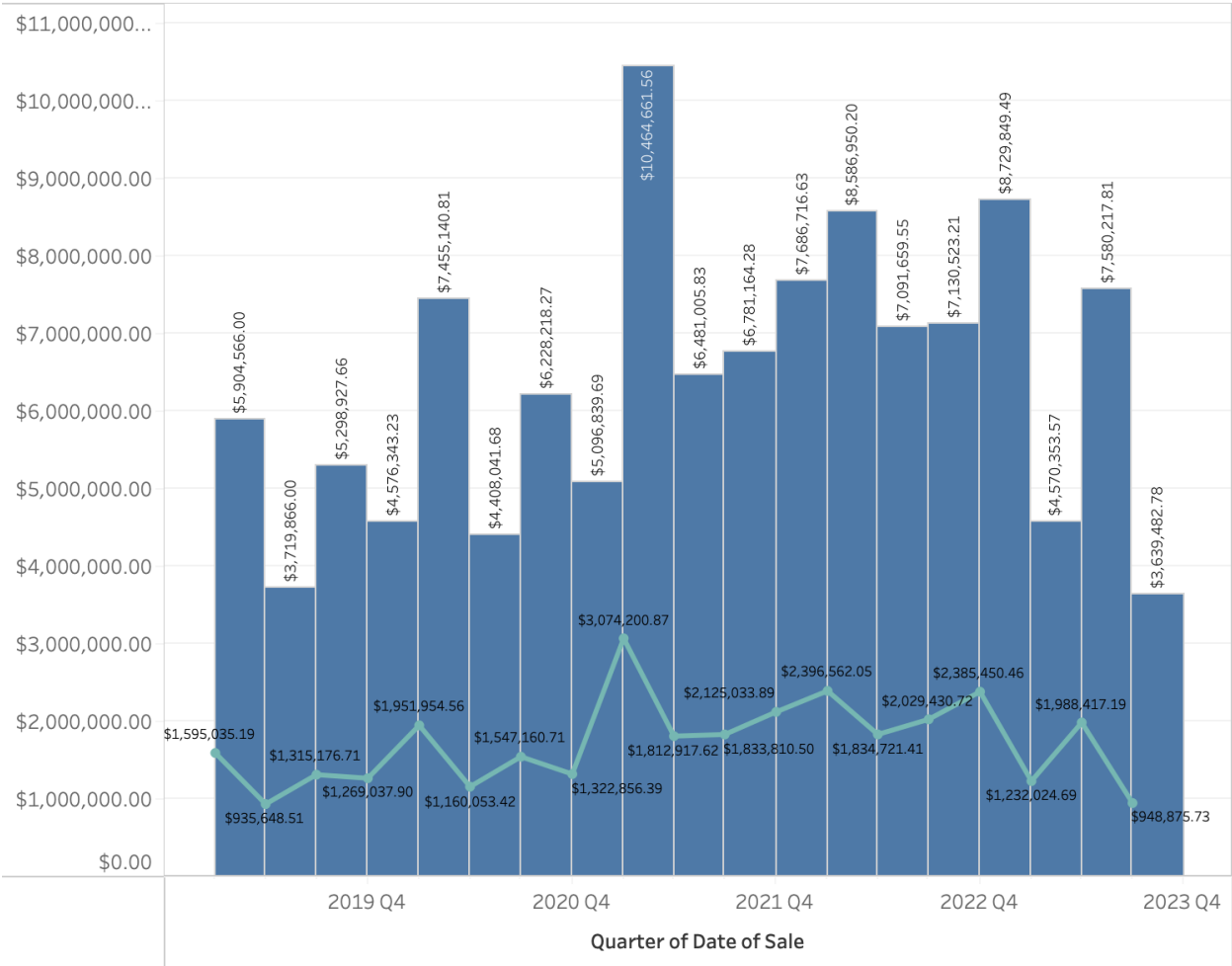
In the evolving landscape of the EV industry, financial metrics serve as a pivotal compass, directing businesses toward sustainable growth and profitability (Hernandez, 2017). Our comprehensive financial analysis seeks to showcase SparkEV's performance across various dimensions. By evaluating aspects such as gross sales distributed by nations, highlighting countries with outstanding gross sales and volumetric sales, and breaking down sales metrics for distinct models, we disclose a comprehensive perspective on the enterprise's financial health.

1. Sales and Revenue growth

From 2019, SparkEV's net sales and gross revenue embarked on an impressive ascent, reaching an unparalleled peak in 2021. This growth, however, witnessed a reversal in fortune by 2023. To put it into perspective, SparkEV's net revenue commenced at \$1.6 million in 2019 and surged to an impressive \$3 million by 2021, but regrettably, this upward trajectory was short-lived. By 2023, the figures receded to \$0.95 million. This ebb and flow in SparkEV's financial

performance highlights the ever-changing dynamics of the EV industry, suggesting the importance of a nimble business approach attuned to market shifts and consumer demands.

Total Quarterly Gross Sales vs Net Revenue by All region(s)



The trends of Gross Sales and Net revenue for Date of Sale Quarter. Color shows details about Gross Sales and Net revenue. The data is filtered on Region, which keeps Asia, Europe and North America.

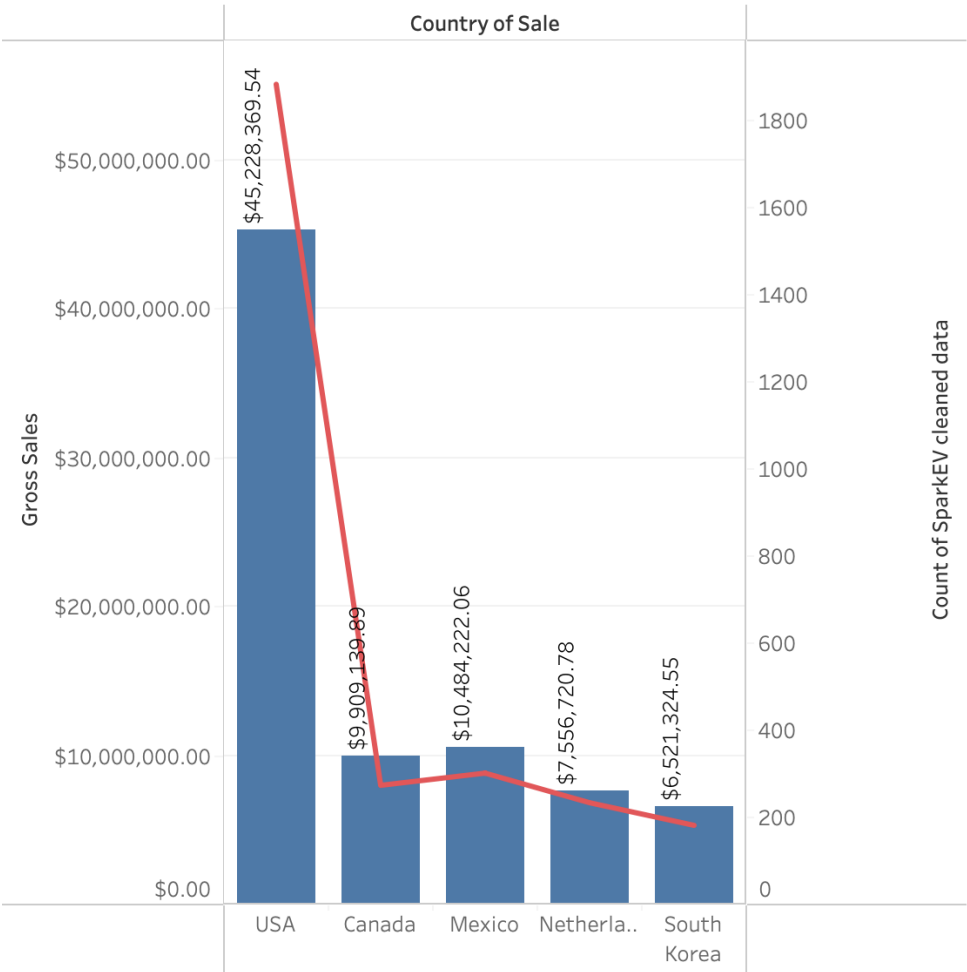
Measure Names
Gross Sales
Net revenue

Figure 2.2. Quarterly Gross Sales and Net Revenue comparison

The EV domain has witnessed substantial growth recently, with different countries showing varied adoption patterns and model choices. The provided figure highlights that gross sales are highest in the USA, with a promising figure of \$45 million, where EV preference is outstanding. This may be due to financial incentives such as tax deductions, which have been shown to boost EV uptake within the country significantly (Jenn et al., 2018). Other countries, such as Canada,

Mexico, the Netherlands, and South Korea, while not matching the US figures, still show a promising trend towards EVs, ranging from approximately 6 million to 9 million recorded in gross sales. This optimistic trajectory can be linked to a combination of government policies, infrastructure development, and public sentiment, all of which have bolstered EV acceptance globally (Jenn et al., 2018).

Top 5 countries based on Net Revenue and theirs Sale Volumes



The trends of Gross Sales and Sale Volume for Country of Sale. Color shows details about Gross Sales and Sale Volume. The data is filtered on Index, which keeps 1, 2, 3, 4 and 5.

- Gross Sales
- Sale Volume

Figure 2.3. Top 5 Countries in Net Revenue and Sale Volumes

2. Models' financial performance

Looking at the financial figures for individual car models, we see some captivating findings about the trend of average gross sales as well as average after-tax profit. From the figure, we can assume that the higher the average gross sales, the higher the average after-tax profits. This phenomenon is frequently traced back to economies of scale, where increased production typically leads to lower unit costs, thereby boosting profit margins (Henderson, 2003). Interestingly, the Storm model, despite having higher average gross sales than the Fertino model, has a lower average after-tax profit (Figure 2.5). Further investigation into the gross sale components of these models reveals that Storm's total variable costs as a percentage of revenue are significantly higher than Fertino's with 56.31% and 50.18% respectively. This suggests that either the Fertino production line is more operationally efficient or that Fertino has more advantageous sourcing deals (Chambers et al., 2006).

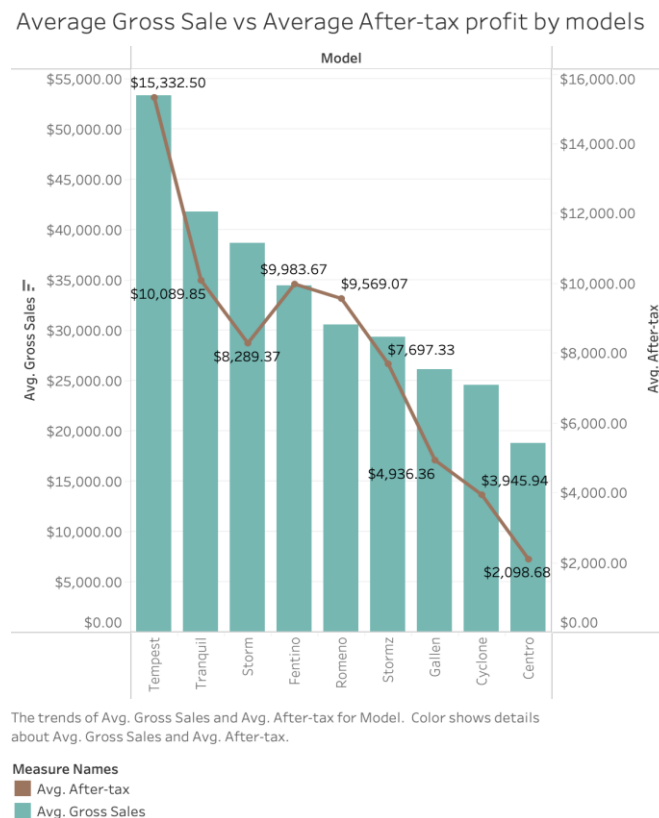


Figure 2.4. Models' Average Gross Sales and After-tax Profit

Gross sale components of Fentino, Gallen, Romeno and 1 more model(s)

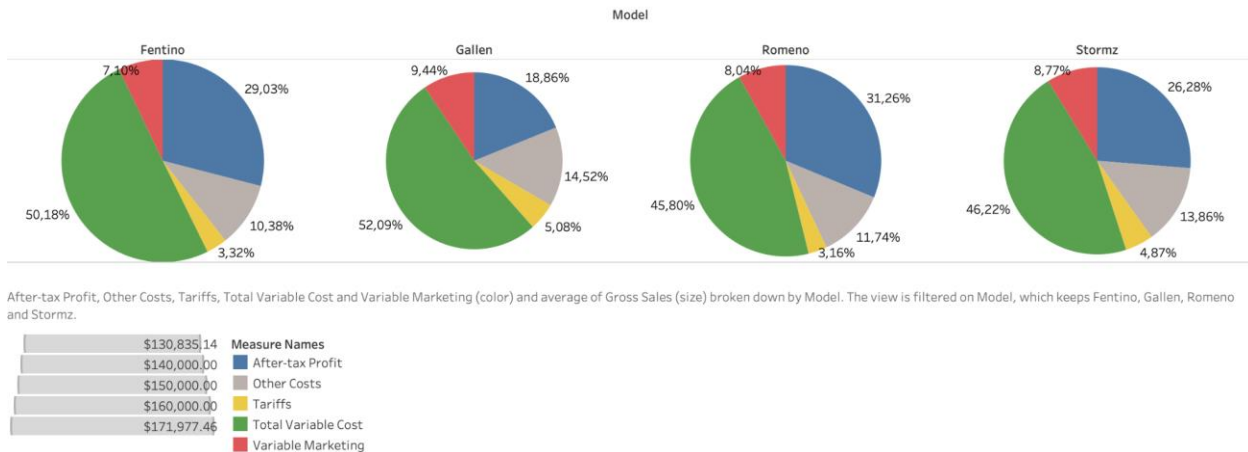
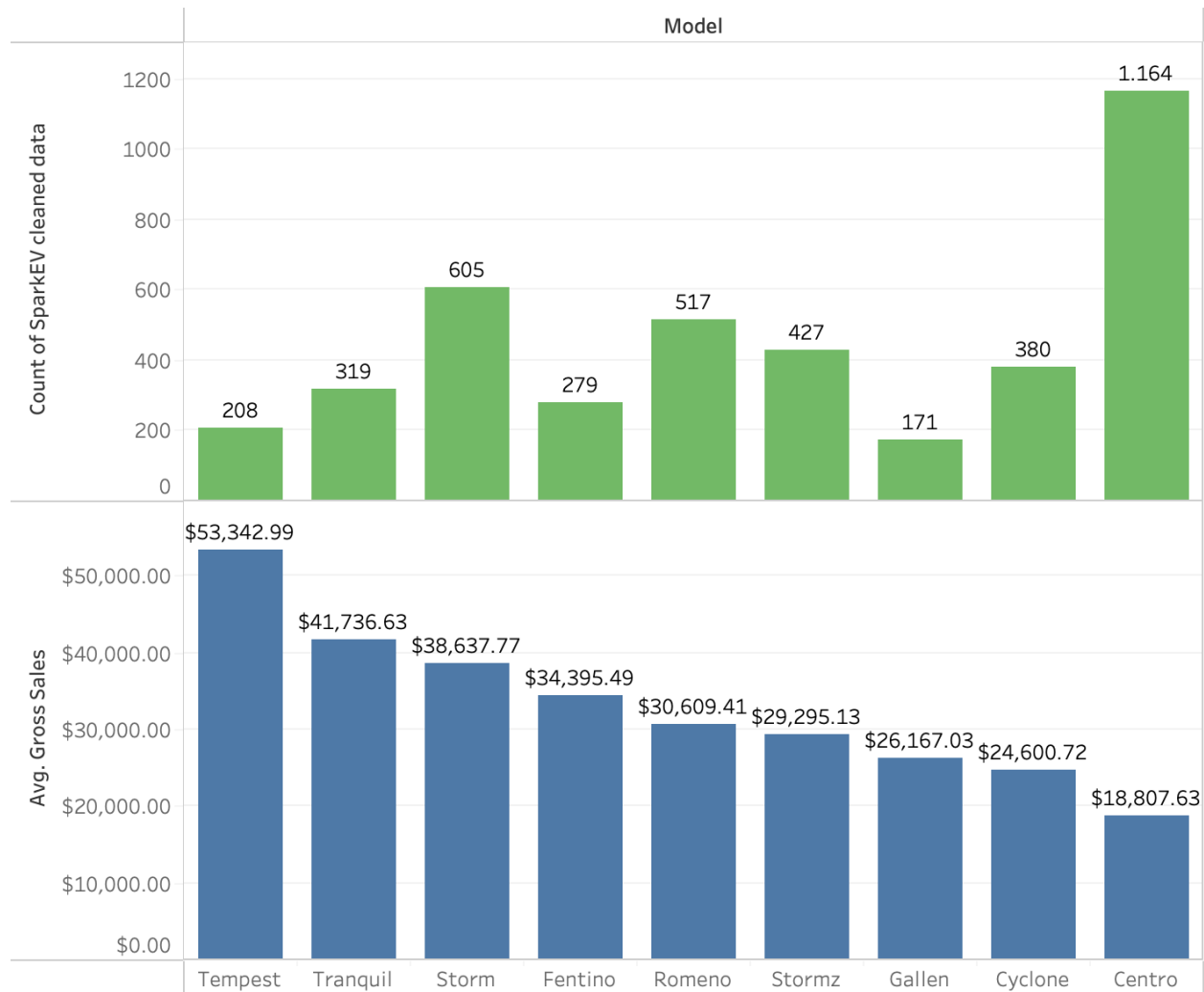


Figure 2.5. *Fentino, Gallen, Romeno, and Stormz's Gross Sales components comparison*

Furthermore, Tempest model stands out with the highest average gross sales of \$53,343 with a matching profit of \$15,333 per sale. With a sales count of 208 units at a much premium rate compared to other brands, the revenue figures soar. This pattern highlights the effectiveness of a high-end pricing strategy in the EV market, which can potentially lead to increased profit margins (Nagle & Muller, 2018). On the other end of the spectrum, Centro records the most negligible average gross sales at \$18,808 per car. Centro, a model that has a significant presence in the USA, with its competitive price tag, probably targets a broader audience, resonating particularly with middle and lower-income groups to help boost the number of sales to 1164 for this model (Nagle & Muller, 2018). This trend has confirmed the importance of adept pricing tactics in influencing EV adoption, especially within economically diverse regions like North America (Alizadeh, 2017).

Sale Volume and Price by model in All

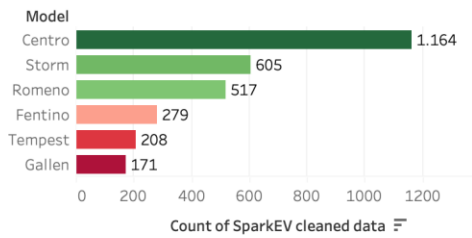


Count of SparkEV cleaned data and average of Gross Sales for each Model. The data is filtered on Region, which keeps Asia, Europe and North America.

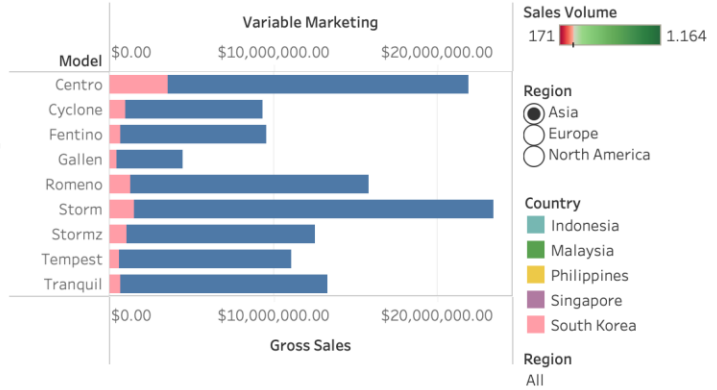
Figure 2.6. Models' Sale Volume and Price

C. Operational Analytics

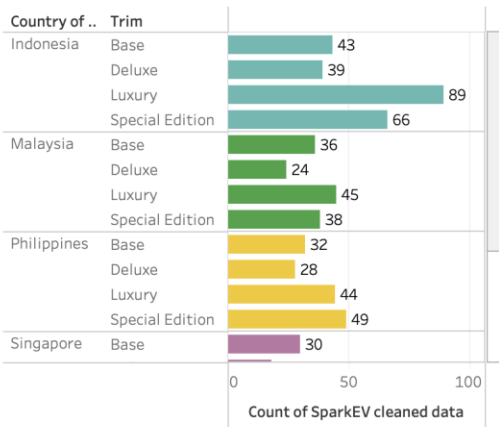
Top 3 and Bottom 3 model by Sale Volume



Variable Marketing Cost / Gross Sales by Models



Favorite trim by each countries in Asia



Days on Lot by Models

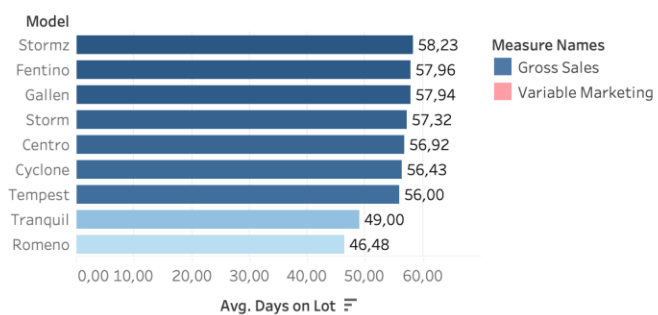


Figure 3.1. *SparkeV Operational Analytics dashboard*

Effective Operations Management is crucial for an enterprise's efficiency and customer satisfaction (Choi et al., 2018). With the assistance of data visualization tools, operational aspects can be analyzed to enhance decision-making. This performance analysis of SparkeV will compare car models, sales, marketing data, and production aspects, emphasizing their significance in supply chain operations.

1. Models' Sales ranking

Top 3 and Bottom 3 model by Sale Volume

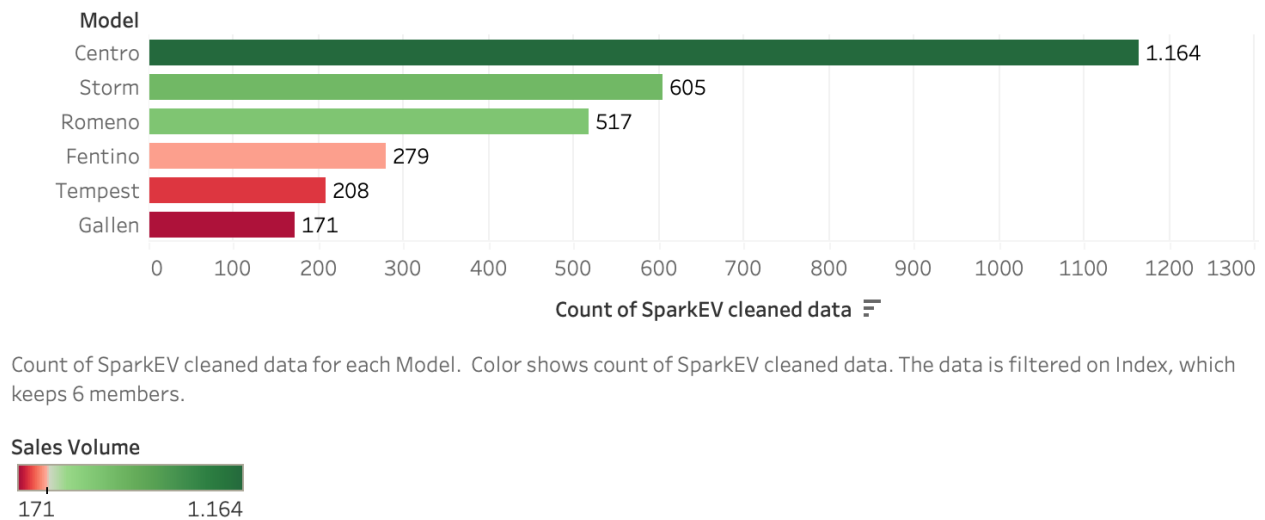


Figure 3.2. *Top 3 and Bottom 3 model by Sales Volume*

The analysis of electric car models shows a clear dominance of the Centro model with 1,164 units sold, nearly double that of the Storm and Romeno models. Despite Storm's high regional sales, SparkEV should prioritize its marketing and stock to meet its demand. Conversely, the less popular models, Fentino, Tempest, and Gallen, with sales significantly lower, suggest a potential reduction in Gallen's availability to shift focus to more demanded models. This sales disparity highlights the necessity for SparkEV to reassess marketing strategies and consider enhancing features for the less popular models.

2. Inventory holding period

Days on Lot by Models

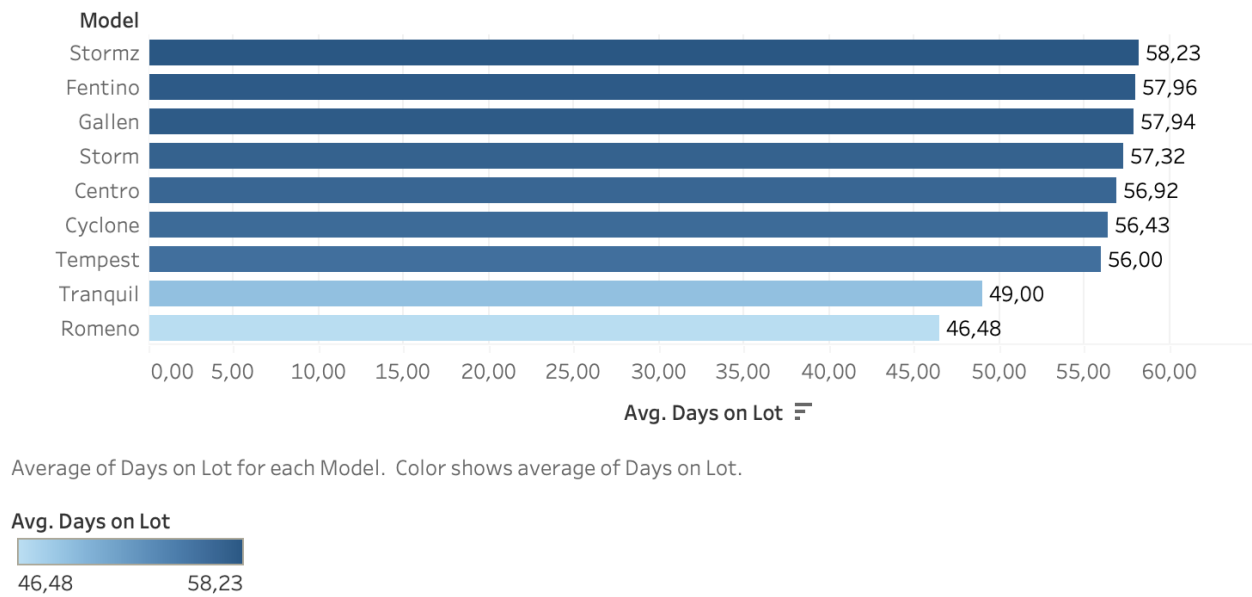


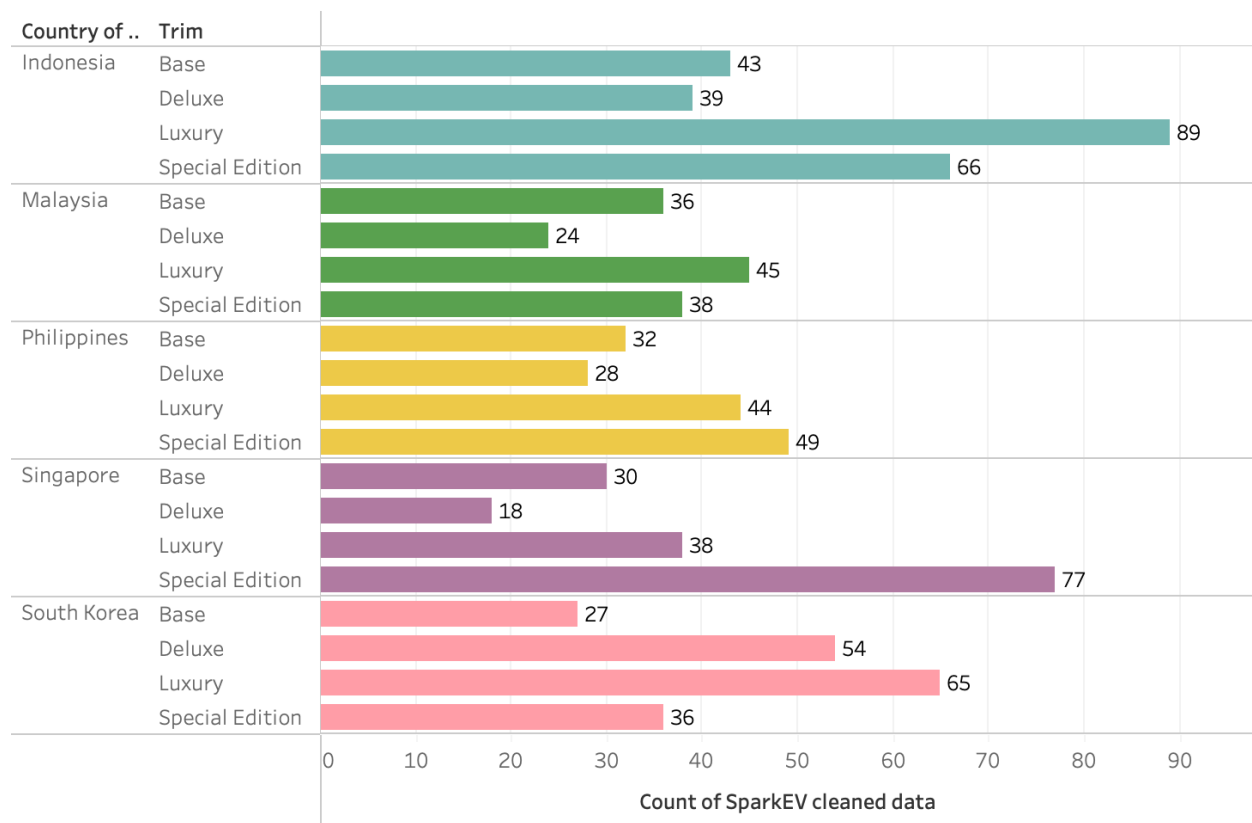
Figure 3.3. *Average Days Models Spend on Lot Prior to Sale*

From a visual comparison detailing the average duration each car model spends on the pre-sale lot, it's apparent that the vehicles, on average, remain unsold for 55 days before being purchased. The Stormz model notably stands out, taking the longest to generate sales, with an average duration of 58.23 days - exceeding the overall average. Conversely, the Gallen model, with a waiting time of 57.94 days, coincidentally boasts the lowest sales figure of just 171 units, as illustrated in Figure 3.1. This may hint that while the Gallen has a niche but dedicated consumer base showing specific interest, its overall demand remains low.

Remarkably, the Romeno model, which is evidently popular among clients, recorded the lowest average days on the lot, taking merely 46 days to find a buyer. Following closely, the Tranquil model took between 48 and 49 days to be sold, which is faster than most. This clearly reflects the high level of enthusiasm and preference buyers have for these particular models. However, it's worth noting that SparkEV might be overstocking the Romeno models. This excess inventory not only escalates storage costs but also potentially hampers the production of other units.

3. Trim preferences and cultural perception

Favorite trim by each countries in Asia



Count of SparkEV cleaned data for each Trim broken down by Country of Sale. Color shows details about Country. The data is filtered on Region, which keeps Asia.



Figure 3.4. *Favorite Trim by each country in Asia*

In Indonesia, the Special Edition's 89 units were more than double the Base's 43 units, hinting at a significant lean towards exclusive editions. Malaysia's Luxury variant, at 45 units, was just a shade above the Base's 36 units, suggesting a relatively balanced market. In the Philippines, the Luxury version's 44 units were slightly higher than the Base, indicating a moderate preference for upscale trims. Singapore's penchant for the Luxury version was evident with its 38 units

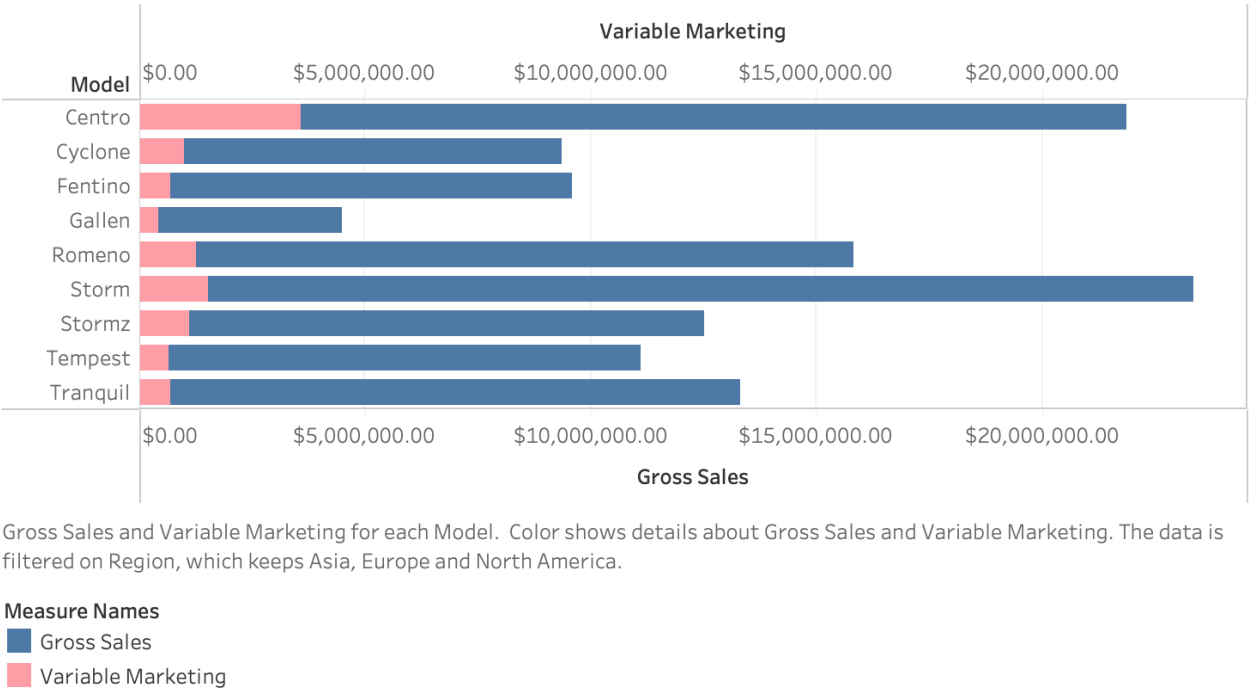
being 26% higher than the Base's 30 units. In South Korea, the Deluxe's 54 units nearly doubled the Base's 27 units, reflecting a clear inclination towards mid-range luxury.

These variations might be attributed to Indonesia's social status desires, Malaysia's income diversity, the Philippines' burgeoning middle class, Singapore's affluent status, and South Korea's search for a luxury-cost balance (Shetty et al., 2020).

In summary, distinct trim preferences across these countries, amplified by stark numerical contrasts, are influenced by socioeconomic and cultural nuances, offering vital insights into companies' market strategies.

4. Variable Marketing Cost and Gross Sales comparison

Variable Marketing Cost / Gross Sales by Models



Gross Sales and Variable Marketing for each Model. Color shows details about Gross Sales and Variable Marketing. The data is filtered on Region, which keeps Asia, Europe and North America.

Figure 3.5. Comparison between Variable Marketing Cost and Gross Sales by Models

From the displayed chart, it is evident how Variable Marketing Expenses correlate with Gross Sales across different vehicle models. Encouragingly, every model yielded positive returns when juxtaposing variable marketing expenses with total sales. Notably, the "Centro" model took the lead in Gross Sales, crossing the \$15 million mark. Despite its average marketing expenses, this

model demonstrates commendable marketing efficiency, generating a substantial profit for the company. It also stands out in EV sales, as seen in Figure 3.1. A deep dive into the North American market reveals that the Centro model reaped a profit of \$19,091,501.10. Nonetheless, its marketing expenses constituted 17.61%, which is relatively high when benchmarked against conventional standards (Marketing, 2023). On the other hand, the "Stormz" model, although bearing high marketing expenses, still showcased impressive sales, particularly in Asia, where it registered \$10,765,524.87. Intriguingly, marketing expenses for this model in Asia were under 10%, indicating the company's adeptness in optimizing expenses for this model. In a nutshell, deciphering the dynamics between variable marketing costs and revenues equips companies to refine their marketing tactics and invest judiciously across vehicle models.

D. Forecast Analytics

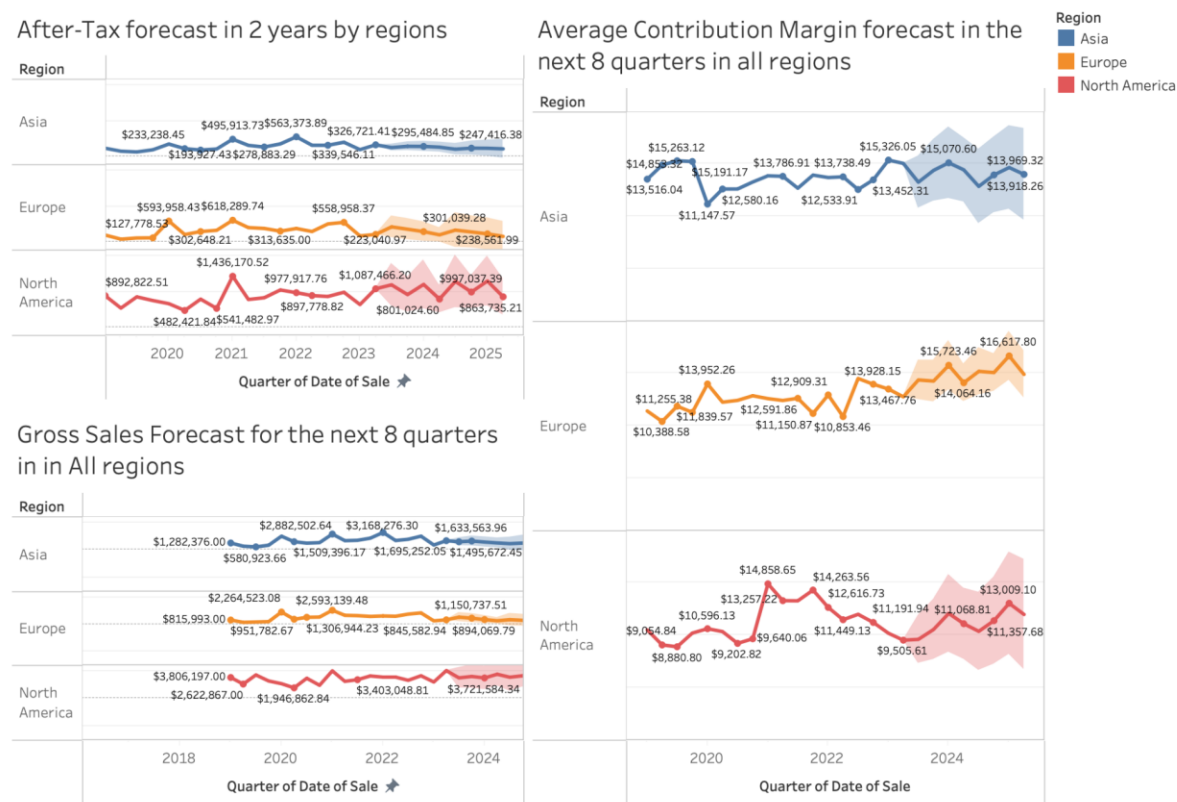
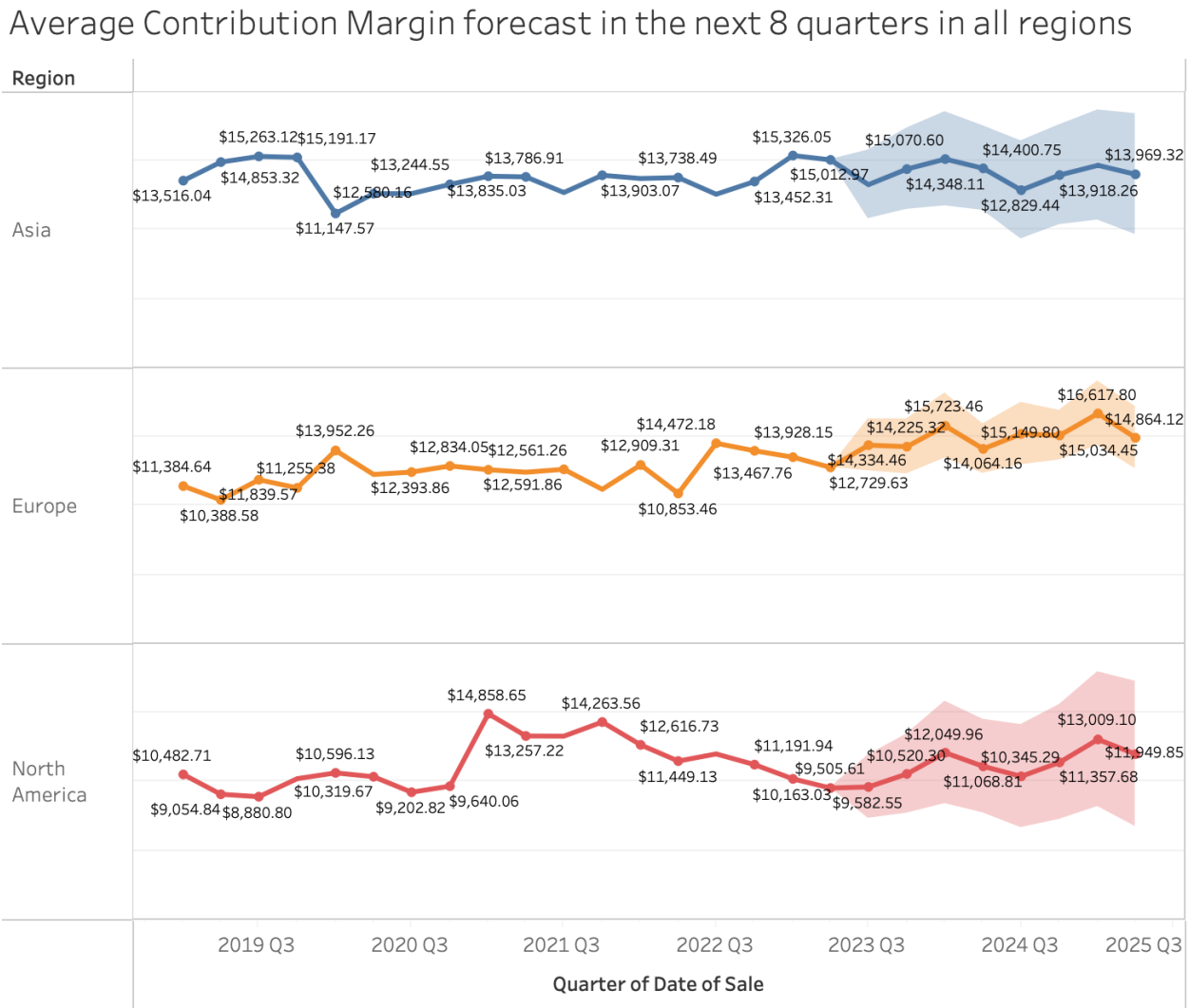


Figure 4.1. SparkeV Forecast Analytics dashboard

Forecasting SparkEV’s performance for the upcoming quarters leading into 2025, it becomes evident that trends in EV sales and contribution margins differ notably across North America, Asia, and Europe.

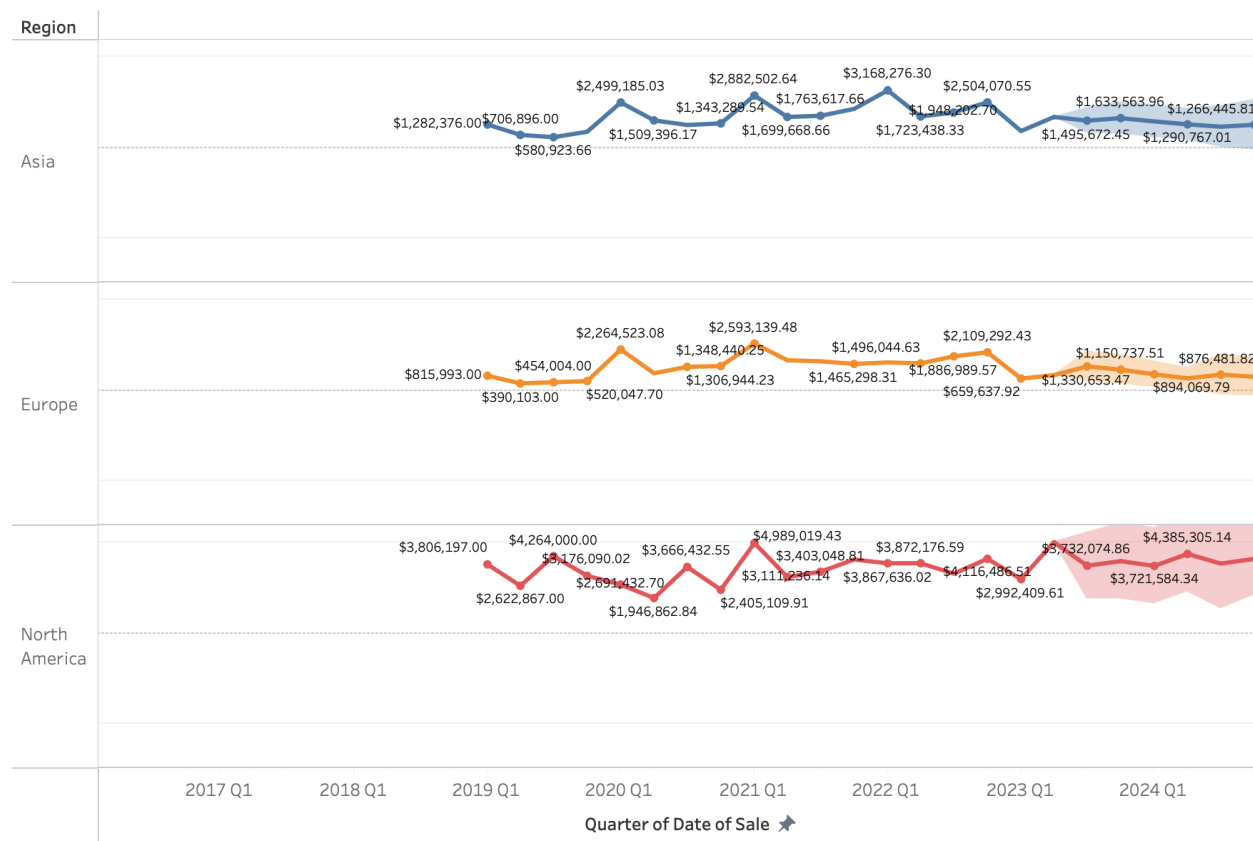


The trend of average of Contribution Margin (actual & forecast) for Date of Sale Quarter broken down by Region. Color shows details about Region.

- Region
- Asia
 - Europe
 - North America

Figure 4.2. Average Contribution Margin forecast in the next 8 quarter per Region

Gross Sales Forecast for the next 8 quarters in in All regions



The trend of sum of Gross Sales (actual & forecast) for Date of Sale Quarter broken down by Region. Color shows details about Region.

Region
■ Asia
■ Europe
■ North America

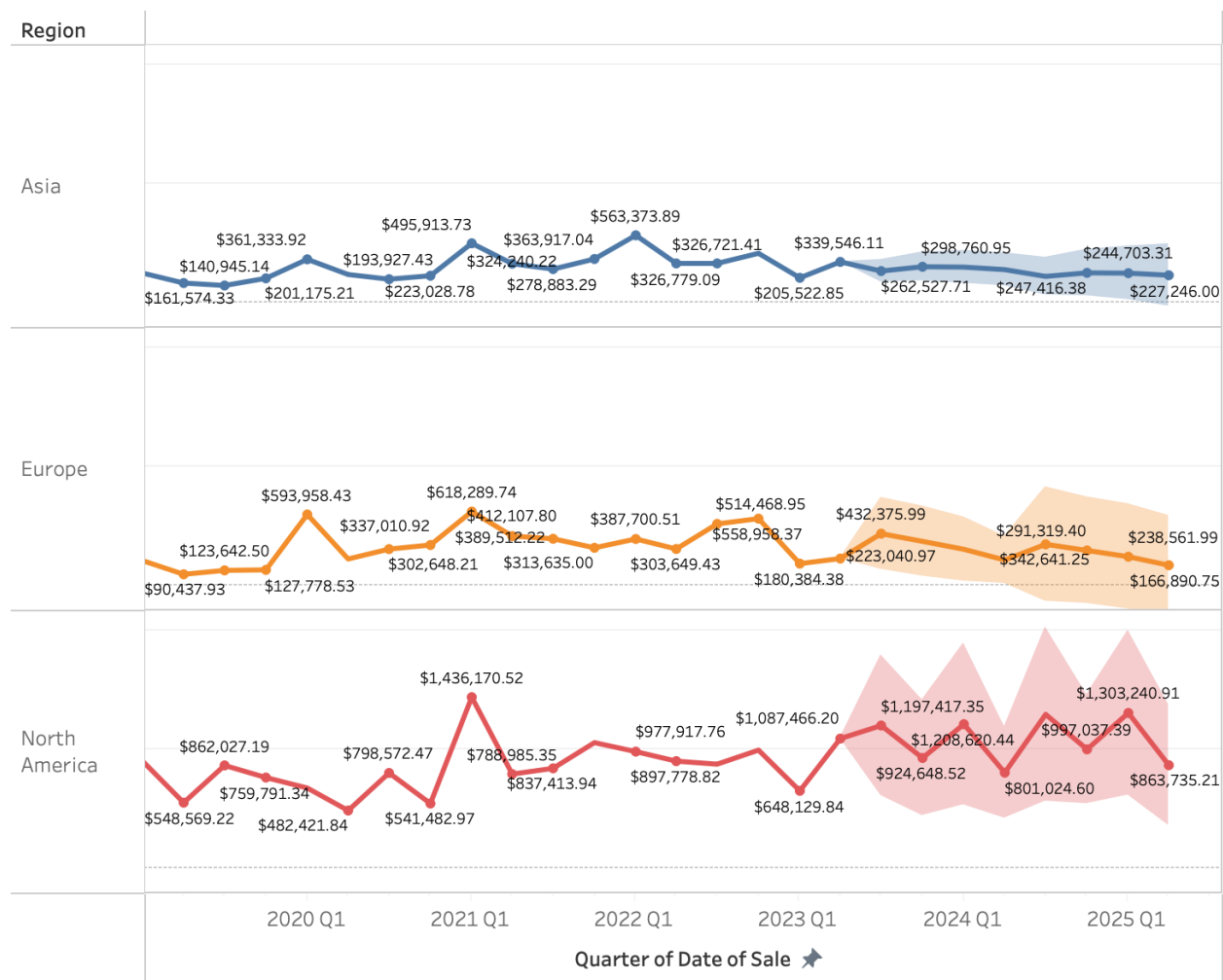
Figure 4.3. Gross Sales Forecast for the next eight quarters in All Regions

According to the Gross Sales forecast, some future trends may be realized.

Region	Forecast Analysis
North America	In North America where SparkEV historically thrived, there's an anticipated decline in sales and contribution margins. According to figure 4.2, Between 2023 and 2025, the North American market is anticipated to experience significant volatility. One possible explanation is market saturation. North America witnessed an early adoption of EVs, and by 2023, many consumers may have already transitioned to EVs (Archsmith et al., 2022). Additionally, there has been a noted rise in competition. Local giants and startups are expanding their EV portfolios, possibly leading to a reduced market share for SparkEV. Recent reports suggest that battery supply chain issues and increased raw material costs could also impact margins in the region (Sun, Hao, et al., 2019).
Europe	Europe is experiencing a renaissance in EV adoption (Münzel et al., 2019). As illustrated in figure 4.2, starting 2019 with only \$0.85 million in gross sales, the region experienced robust growth with projections indicating a rise to more than \$2 million in the first quarter of 2023. Governmental incentives, combined with strict emission regulations, are driving consumers towards EVs (Santos & Davies, 2019). Moreover, European nations are heavily investing in EV infrastructure, making it easier for consumers to make the switch. This is backed by data showing a 15% year-on-year increase in the number of public charging stations across major European cities (Münzel et al., 2019).
Asia	Asia's trajectory is more complex. According to figure 4.3, the market initially showed promise, peaking in 2022 quarter 1 at \$3 million, but then depicted fluctuations. One contributing factor could be the diverse market conditions within the continent. While nations like Japan and South Korea are seeing rapid EV adoption due to technological advancements and government backing, others face challenges (Kim et al., 2019). For instance, in some densely populated cities, the lack of infrastructure for personal vehicle charging could be a barrier (Knez et al., 2019). Moreover, local competitors are emerging, providing vehicles tailored to specific regional needs, which could be impacting SparkEV's market share (Singh Patyal et al., 2021).

Table 2. Forecast analysis per Region

After-Tax forecast in 2 years by regions



The trend of sum of After-tax (actual & forecast) for Date of Sale Quarter broken down by Region. Color shows details about Region.

Region
 Asia
 Europe
 North America

Figure 4.4. Profit After-Tax forecast in 2 years by Regions

Considering the after-tax profit figures, these trends become more palpable. North America's decrease could be attributed to the competition mentioned above, but also possibly due to potential tariff increases or changes in trade agreements affecting imported vehicle components (Helveston, 2021). Meanwhile, Europe's growth trajectory suggests that SparkEV should consider further investments there, perhaps in localized research and development or even partnerships with local entities (Santos & Davies, 2019).

In conclusion, for SparkEV to navigate the approaching years successfully, a multifaceted approach is essential. This includes not only enhancing product offerings but also fortifying supply chains, understanding regional nuances in-depth, and possibly exploring collaborations in high-growth areas.

E. Recommendations

From the data analysis, forecasts, and subsequent insights, some recommendations for SparkEV and their impacts on the organization can be found in Table 2. To appropriately apply these recommendations, further research and analysis should be conducted to quantify available space for improvement.

Segment	Insights: Challenges and Opportunities	Recommendations	Impacts
Market share	- North America's large consumption and US market dominance - Market on the upward trajectory, however, is relatively saturated.	- Enhance marketing, establish strategic local partnerships, and elevate the customer service experience (Ou et al., 2019; Knez et al., 2019). - Conduct comprehensive regional analyses and replicate success factors from North America.	Understanding the unique dynamics behind the U.S.'s stellar performance can pave the way for similar triumphs elsewhere (Knez et al., 2019). Replicating these success stories in other markets could extend SparkEV's global footprint.
	Europe and Asia's growth potential - Favorable government incentives (tax reductions, tariff decrease) to increase EV usage. - Tax benefits in Europe offer a platform to boost tax revenue (Münzel et al., 2019). - Asia's large and	- Tap into underexploited markets with tailored marketing campaigns that align with regional socio-economic nuances (Rehman et al., 2020). - Conduct market research to better understand the specific needs and preferences of consumers in	Leveraging favorable government incentives in Europe can lead to increased EV adoption, expanding SparkEV's market presence in the region. This expansion may result in higher sales and revenue, thus positively impacting the company's financial health.

	growing population	different regions. This information can be used to develop new vehicles and marketing campaigns that are more likely to resonate with consumers. - Educational workshops, marketing campaigns, and other outreach initiatives to promote the adoption of EVs. - Partner with local governments and businesses to promote the adoption of EVs	
	China is a particularly promising market, given its leading EV adoption and growing urban population (Ou et al., 2019).	Forge strategic local partnerships and tailor its approach to China's unique commercial ecosystem and consumer preferences (Li et al., 2020).	By deeply rooting its operations within China's dynamic landscape, SparkEV could significantly amplify its global market share, harnessing the region's thriving momentum for sustainable automotive solutions.
	Global factors may cause declining sales forecast. - Global chip shortage - Rising material costs - Russia - Ukraine war - Economic downturn (IEA, 2023)	- Diversify supply chain sources to mitigate the risks of components shortages. Larger players have considered self-supply, but this is not yet a viable solution for SparkEV's size. - Cost optimization strategies should be a focus given the rising material costs. This can be renegotiating supplier contracts, researching alternative	- Supply chain and financial resilience ensure continued production and operation despite disruptions or challenges. - Cost optimizations maintain profit margins, supporting financial health and profitability (Stanton, 2023).

		materials, or implementing a more efficient manufacturing process. - Financially, maintaining a healthy cash reserve and evaluating capital allocation strategy can ensure financial resilience against economic uncertainties (Bachmann et al., 2013).	
Product profile	Storm leads in sales while Tempest demonstrates operational excellence -> This suggests potential cost-saving opportunities for the Storm. Lessons from the Tempest's success can be applied for overall efficiency.	The Storm model requires resource reallocation for improved cost-revenue balance	Resource management is essential to production and operation efficiency. Stable growth can be achieved with vigilant resource utilization (Stanton, 2023).
	There is significant variation in individual model performance and popularity within each segment. - The Tempest's success in the luxury segment indicates an appetite for premium EVs, opening opportunities for SparkEV in the high-end niche. - Centro's lower price and accessibility contribute to its large sale volume. However, its profit margin is low.	- Given the saturated market, Tempest should regularly be repositioned to align better with its market appeal and price point, maintaining its performance. - The strategic direction should emphasize market entry, product optimization, and prudent financial decisions. - Models like Centro should be fine-tuned to broaden their appeal	- By recognizing the appetite for premium EVs within individual buyers, SparkEV can seize opportunities in the high-end niche, potentially increasing sales revenue, and market share. - Regular repositioning of models ensures their continued appeal and success in the luxury segment, showcasing adaptability to market dynamics and consumers' changing preferences. - Addressing Centro's low profit margin by

		without compromising profitability.	fine-tuning the model helps improve profitability while maintaining its accessibility and sales volume.
	Cyclone and Fentino discontinuations after poor performance.	Management should regularly optimize the model portfolio by reassessing marketing strategies, making product enhancements, or phasing out underperforming models (Stanton, 2023).	- Reviewing setbacks offers valuable insights into avoiding future obstacles related to supply chain or production, ensuring market responsiveness, sustainability and building brand image. - Regularly optimizing the model portfolio and reassessing marketing strategies contribute to operational efficiency. By identifying and prioritizing high-performing models and marketing strategies, SparkEV can allocate resources more effectively, reducing wastage and maximizing profitability.
Distribution	Sales through dealers are consistently the lowest among regions, leaving room for improvement.	Establish and maintain strong dealer network: - Recruit new dealers. - Establish training programs. - Invest in dealer facilities	- Increase competitiveness against other automakers. - Customer leveraging can lead to long term partnerships and consistent sales, leading to a stable revenue stream.
	- Retail sales bring in the most profit, yet volume is lower than industry	Enhance the retail experience to appeal to customers: - Improving customer	- Retail sales already bring in the most profit, and catering to customer needs and

	average, wasting potential earnings. - Regional consumer preferences are different due to cultural and economic factors. E.g., Singaporeans have a heavy preference for luxury products, while South Koreans seek for a luxury-cost balance.	service - Offer competitive pricing, done through careful financial and regional market analysis. - Provide variety of financing options.	preferences can further optimize earnings and sales. - Adapting to regional consumer preferences allows SparcEV to capture a larger market share. By offering what customers in different regions desire, the company can solidify its presence in various markets, reinforcing its competitive position on a global scale. - Premium retail experience fosters greater brand loyalty and image.
	- Fleet sales dominate in volume, signifying a strong corporate customer base. - A niche market is needed to solidify market share in a saturated industry. - Dominant markets are saturated with large industry players, yielding a need for differentiation.	Partner with fleet management companies and other businesses to promote the sale of SparcEV vehicles through the fleet sales channel. Or Consider offering fleet management services to cater to the largest customer segment.	- The focus can increase market share within the specific customer segment.
	Days on lot are relatively long	Focus efforts on implementing a more efficient inventory management system to reduce the average time spent on the pre-sale lot (Erkayman, 2019).	- Reduced idling leads to lower storage costs, security, and maintenance. - Reduce the capital tied to unsold vehicles by improving product turnover.

Table 3. Business recommendations from insights and their impacts

V. Conclusion

This analysis of SparkEV's data from 2019 to 2023 provides essential insights and recommendations for the company's future in the electric vehicle industry. As a small company in the automotive manufacturing industry, SparkEV needs to pay special attention to data management for maintenance of high-quality data integrity. Privacy and security risks must be addressed to ensure reliable decision-making.

SparkEV's dominant market in North America and the growing trajectory suggest the importance of strengthening its foothold in this region by elevating its marketing initiatives, forging strong local partnerships, and amplifying its customer service. In parallel, the company should pivot towards Europe and Asia, not just by increasing its market presence but by tailoring strategies that cater to the unique demands of these regions. Nevertheless, the successes and best practices that have propelled growth in North America should not be abandoned but rather adapted and replicated appropriately across these new markets to ensure a harmonized global approach.

Financial analysis highlights the need for an adaptable business approach in response to market dynamics. Tailoring pricing strategies to market segments can optimize profitability. Operational insights emphasize the need to optimize the product portfolio. Learning from the Tempest model's operational excellence can improve the cost-revenue balance. Repositioning the Tempest model in line with market appeal and pricing is advisable. Diversifying the product range, conducting market research, and implementing cost management strategies will appeal to a broader demographic and enhance profitability. Sales and distribution channels should be improved by enhancing the retail experience, aligning with regional preferences, and strengthening the dealer network.

In summary, SparkEV's journey in the electric vehicle industry necessitates an agile approach involving marketing, production, product optimization, and financial management. With further analysis, the collaboration with local partners, market research, and continued investments in research and development will solidify SparkEV's position as a leader in the global electric vehicle industry.

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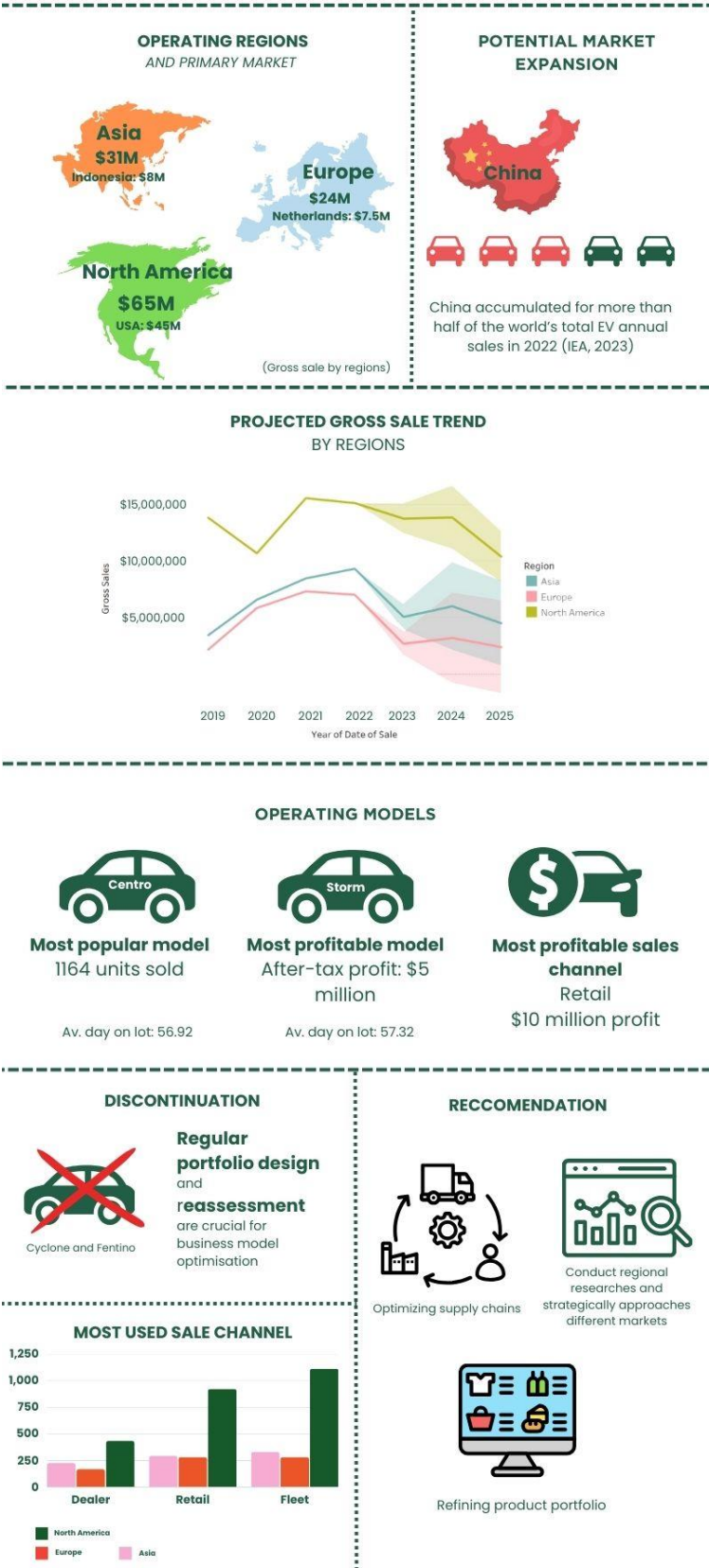
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SPARK ELECTRIC VEHICLE
SALE ANALYSIS



The analysis has been consolidated and condensed into a single infographic, encapsulating the key insights into SparkEV's performance.